

Two-Factor RFID + Face Recognition Door

Master Engineering Package - Requirements, Specification & Design Baseline

Document ID	DOC-00	Status	Implementation Baseline
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Purpose: One coherent reference for building the MVP without overengineering it. DOC-01 is normative for requirements; DOC-02 is the architectural model; DOC-03 is the implementation contract.

1. Executive Summary

This project demonstrates physical access control using two independent factors: possession of an enrolled RFID card and presentation of the corresponding enrolled face. The ESP32/MFRC522 handles device-side sensing and actuation. The laptop handles face recognition, the 10-second session window, authorization policy, persistence, and logging.

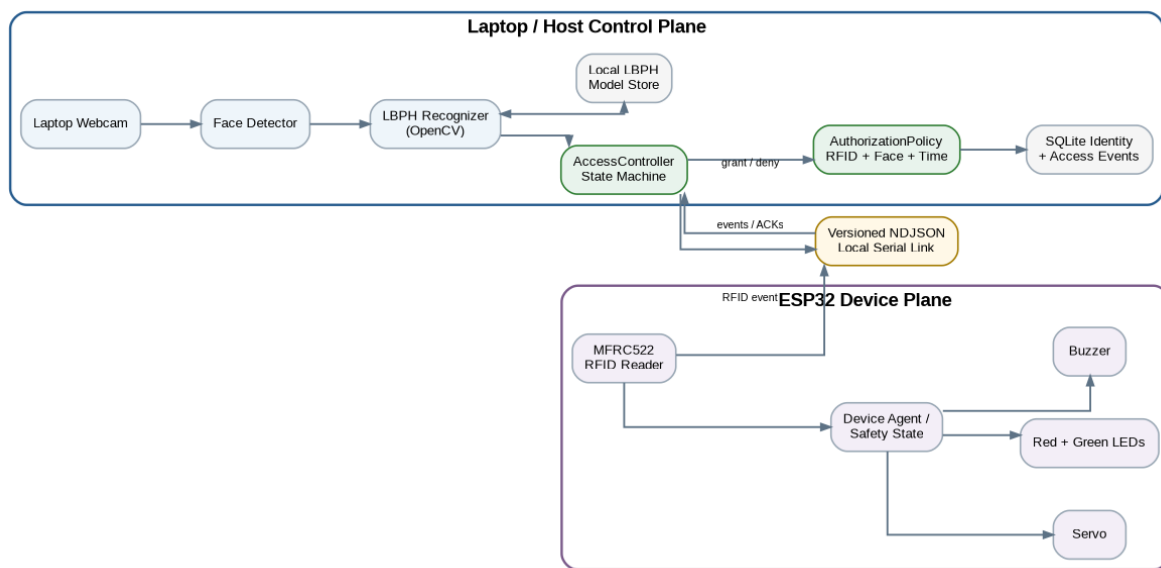


Figure 1. System architecture

2. System Contract

Property	Canonical value
Factors	RFID card + matching face
RFID component	ESP32 + MFRC522 + cards
Vision	Laptop webcam + OpenCV LBPH
Verification window	10 seconds after valid RFID

Property	Canonical value
Positive output	Servo opens + green LED + short beep
Negative output	Servo remains closed + red LED + long buzz
Primary user population	Small local set; typically 1-2 enrolled people for MVP demonstration
Persistence	SQLite + local face/model files
Host-device link	Versioned NDJSON over local USB serial
Security posture	Prototype; fail closed; no liveness; RFID UID not cryptographic credential

3. Canonical Decision

GRANT iff:

card is active and enrolled
AND now < session.deadline
AND exactly one usable face is visible
AND LBPH distance <= configured threshold
AND recognized identity == card owner
AND stable match count >= configured N

Otherwise:

DENY / TIMEOUT / FAULT
-> servo closed

4. Documents in This Package

Document	Use
DOC-01-SRS	Requirements, acceptance criteria, security limitations, traceability.
DOC-02-HLD	Components, boundaries, data ownership, failure handling, scalability seams.
DOC-03-LLD	Modules, interfaces, protocol, pin baseline, pseudocode, implementation constraints, test pyramid.

5. Build Sequence

1. Bring up ESP32 + MFRC522 and prove stable card UID reads.
2. Bring up servo, LEDs, and buzzer with a hardware-only device test.

3. Implement versioned serial protocol and a device simulator.
4. Implement SQLite identity store and RFID enrollment.
5. Implement webcam capture and face detection.
6. Implement LBPH enrollment and model persistence.
7. Implement AccessSession and AuthorizationPolicy as pure/testable code.
8. Integrate real hardware through adapters.
9. Calibrate the LBPH threshold using actual enrolled samples and the actual webcam.
10. Execute ACC-001..ACC-010 and record evidence in the test log.

6. Scalability Strategy

The design is scalable at the architecture boundary level but intentionally small at the deployment level. The core policy never depends on SQLite, a specific camera, a specific recognizer, or a specific device link. That permits later evolution to multiple doors, stronger card technologies, better vision models, centralized persistence, or networked device gateways without rewriting the access-policy core.

Today:

Host -> Serial -> One ESP32 -> One door

Later, if justified:

Host Policy Core

-> Device Gateway -> Door Registry -> N devices

or

Policy Core -> Repository implementation -> Central store

or

Policy Core -> FaceRecognizerPort -> stronger recognizer

7. Explicit Non-Optimization Rule

Do not optimize for people who are not here: The first implementation should be easy to read, easy to run, and easy to test on one laptop and one door. Capacity for future growth is provided by interfaces and stable contracts, not by unused infrastructure.

8. Security and Honesty

- The project proves two-factor identity binding; it does not prove production-grade biometric security.
- Printed-photo attacks may bypass LBPH because no liveness/presentation-attack detection is implemented.
- RFID UID-based identification is cloneable and is not equivalent to secure smart-card cryptography.
- The physical door/servo mechanism, enclosure, power, and manual bypass are outside the software assurance boundary.
- The safest software behavior on uncertainty is always closed door, not convenience.

9. Final Acceptance Gate

MVP passes: A live demonstration can show: correct card + correct face -> open; correct card + wrong face -> closed; unknown card -> closed; no face/slow face -> closed; communication or camera failure -> closed; and

the known printed-photo limitation is documented.

10. Reference Sources

OpenCV 4.13 LBPHFaceRecognizer API:

https://docs.opencv.org/4.13.0/df/d25/classcv_1_1face_1_1LBPHFaceRecognizer.html

NXP MFRC522 product page: <https://www.nxp.com/products/MFRC52202HN1>

NXP MFRC522 data sheet: <https://www.nxp.com/docs/en/data-sheet/MFRC522.pdf>

Espressif ESP32 UART documentation: <https://docs.espressif.com/projects/esp-idf/en/release-v5.2/esp32/api-reference/peripherals/uart.html>